

Eduarn-like LMS Full Setup Guide

1. Project Structure

Create main folder: lms-project with backend and frontend directories.

2. Backend Setup

Initialize Node.js project and install dependencies:

```
npm init -y
npm install express mongoose cors bcrypt jsonwebtoken multer
```

Backend Folder Structure

```
backend/
  ├── models/
  │   ├── User.js
  │   └── Course.js
  ├── routes/
  │   ├── auth.js
  │   └── course.js
  └── server.js
```

server.js

```
const express = require("express");
const mongoose = require("mongoose");
const cors = require("cors");

const app = express();
app.use(cors());
app.use(express.json());

mongoose.connect("mongodb://127.0.0.1:27017/lms");

app.use("/auth", require("./routes/auth"));
app.use("/courses", require("./routes/course"));

app.listen(5000, () => console.log("Server running on 5000"));
```

User Model

```
const mongoose = require("mongoose");

const userSchema = new mongoose.Schema({
  name: String,
  email: String,
  password: String,
  role: { type: String, enum: ["student", "instructor", "admin"] }
});
```

```
module.exports = mongoose.model("User", userSchema);
```

Course Model

```
const mongoose = require("mongoose");

const courseSchema = new mongoose.Schema({
  title: String,
  description: String,
  price: Number,
  instructor: String,
  videos: [String]
});

module.exports = mongoose.model("Course", courseSchema);
```

3. Frontend Setup

Create React app and install dependencies:

```
npx create-react-app frontend
cd frontend
npm install axios react-router-dom
```

Courses Page Example

```
import { useEffect, useState } from "react";
import axios from "axios";

export default function Courses() {
  const [courses, setCourses] = useState([]);

  useEffect(() => {
    axios.get("http://localhost:5000/courses")
      .then(res => setCourses(res.data));
  }, []);

  return (
    <div>
      <h1>Courses</h1>
      {courses.map(c => (
        <div key={c._id}>
          <h3>{c.title}</h3>
          <p>{c.description}</p>
          <p>■{c.price}</p>
        </div>
      ))}
    </div>
  );
}
```

4. Run Project

Backend: node server.js

Frontend: npm start

5. Test API

Use Postman to create course:

```
POST http://localhost:5000/courses/create
```

```
{  
  "title": "React Course",  
  "description": "Learn React",  
  "price": 999,  
  "instructor": "Admin"  
}
```

6. Next Steps

Add authentication UI, video upload, payment integration, dashboards, deployment.